

WorkshopPlus: Azure AI Platform and Services

RAG Patterns: GraphRAG and the GraphRAG Accelerator

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Learning Units covered in this Module



Intro to GraphRAG



The GraphRAG Library



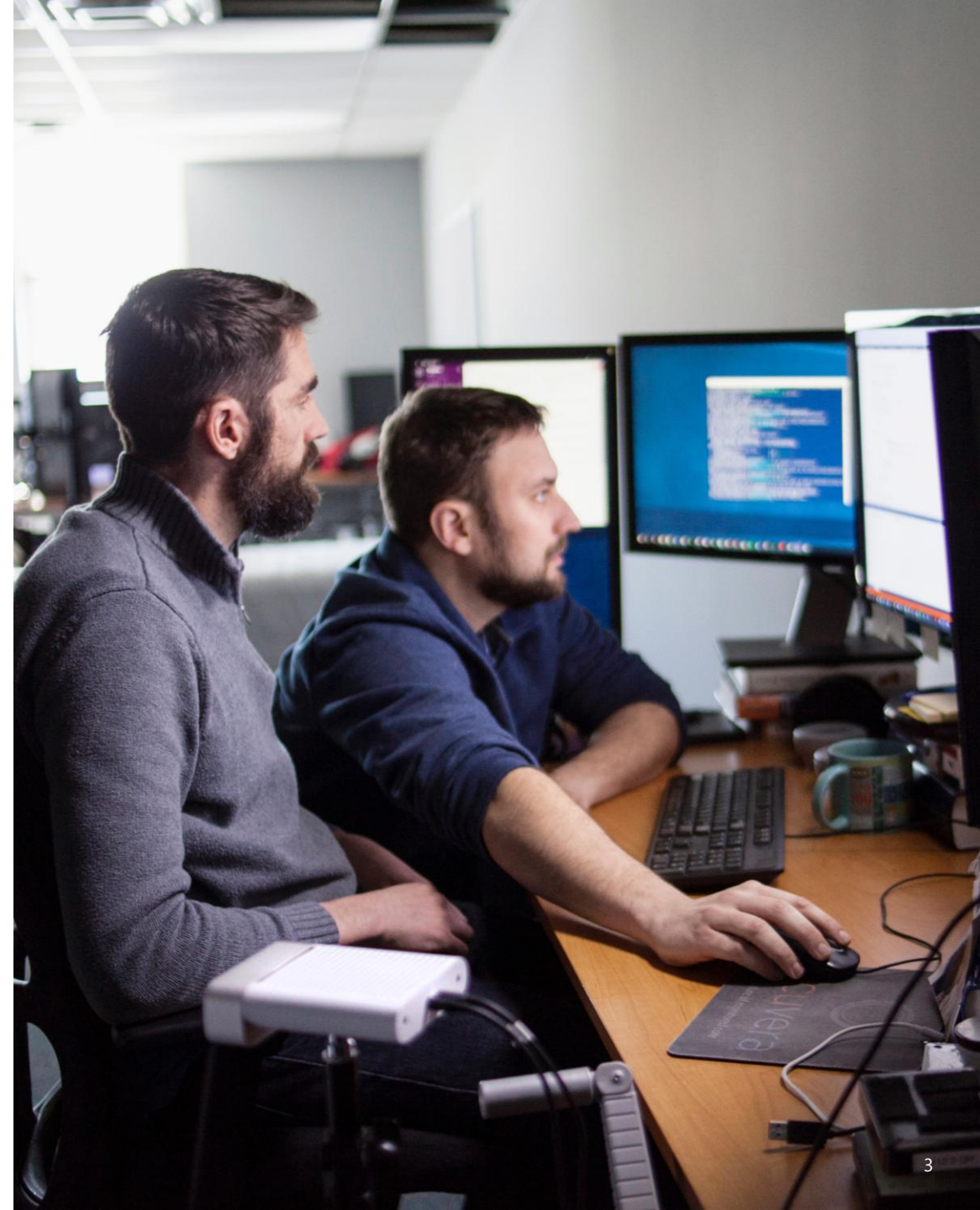
Augmenting with Domain Expertise



Exploring the Knowledge Graph



The GraphRAG Accelerator



Objectives

After completing this Learning, you will be able to:

1

Understand the fundamentals of GraphRAG

- ✓ Learn what GraphRAG is and how it fits into Retrieval-Augmented Generation (RAG) patterns.
- ✓ Explore the core components and architecture of the GraphRAG library.

2

Understand how to augment GraphRAG with domain expertise

- ✓ Discover how to integrate domain-specific knowledge into AI workflows using GraphRAG.
- ✓ Examine techniques for enriching knowledge graphs with curated data sources.

3

Understand the structure and utility of knowledge graphs

- ✓ Analyze how knowledge graphs are constructed and queried within the GraphRAG.
- ✓ Identify use cases where knowledge graphs enhance AI reasoning and retrieval.

4

Understand the GraphRAG Accelerator and its capabilities

- ✓ Get familiar with the tools and templates provided in the GraphRAG Accelerator.
- ✓ Learn how to deploy and customize the accelerator for enterprise scenarios.

5

Understand how to apply GraphRAG in practical scenarios

- ✓ Participate in hands-on labs and demonstrations to reinforce learning.
- ✓ Evaluate real-world applications and performance considerations of GraphRAG.



An Introduction to GraphRAG

What does it solve?

How does it work?

What is available now?

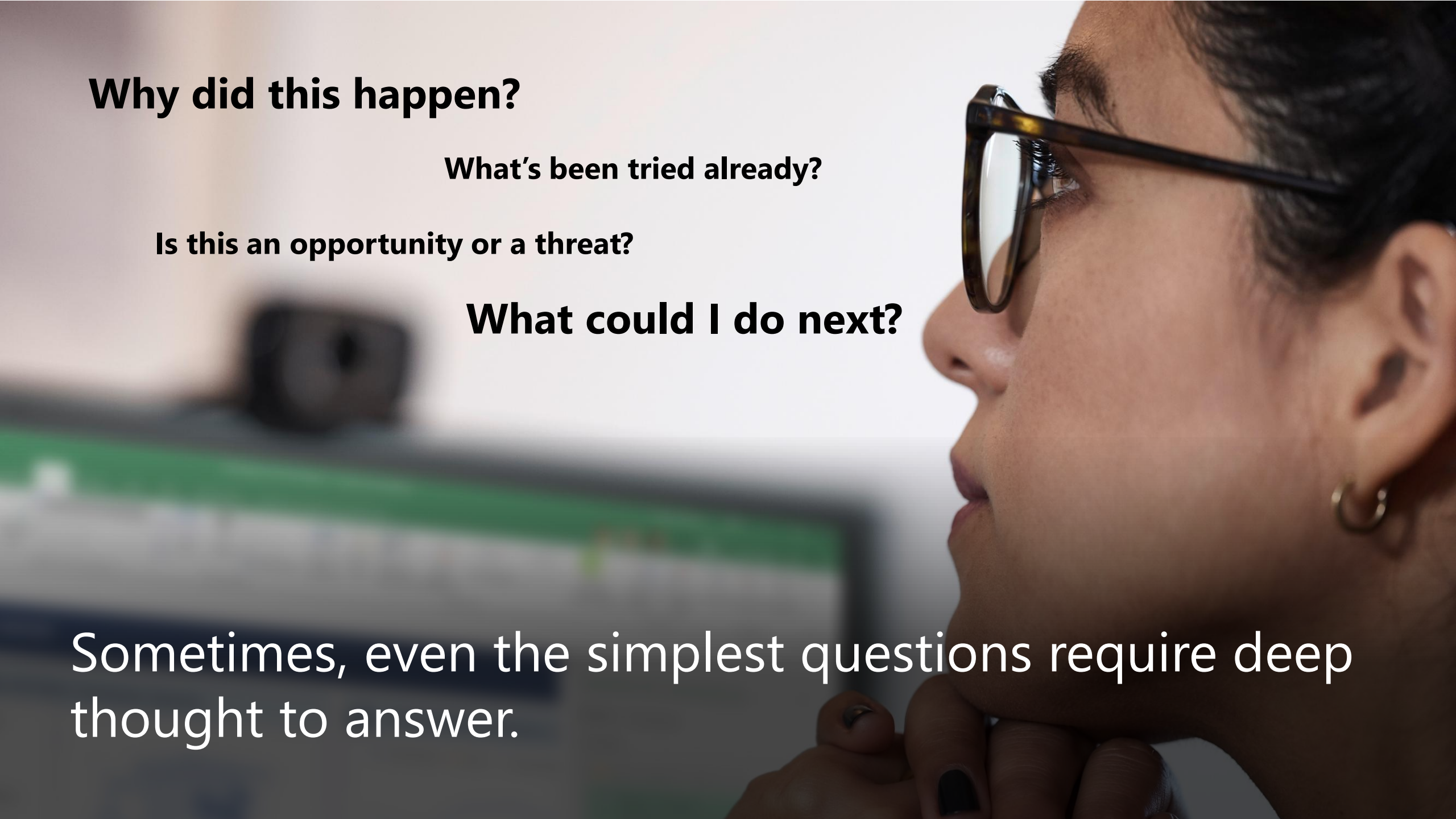
Why did this happen?

What's been tried already?

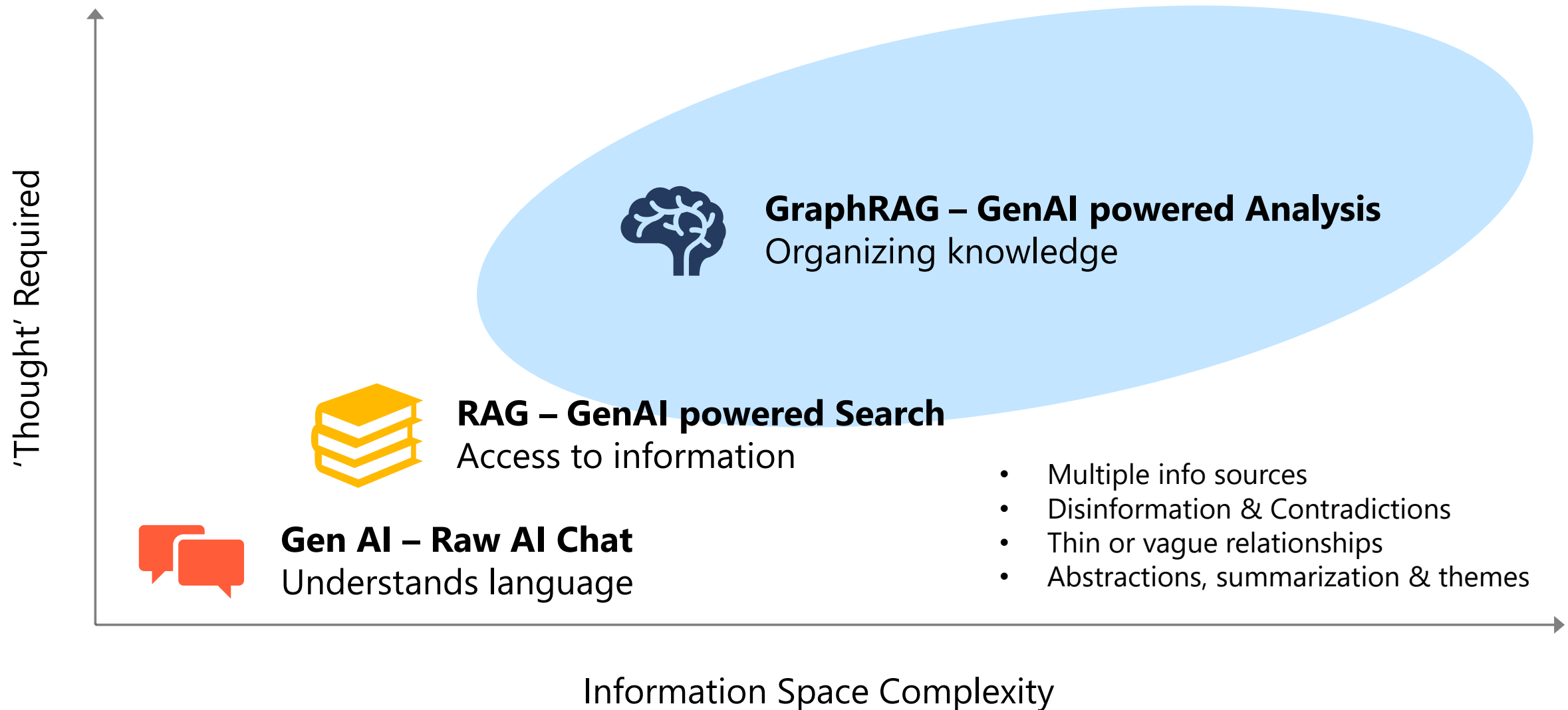
Is this an opportunity or a threat?

What could I do next?

Sometimes, even the simplest questions require deep thought to answer.



As information complexity grows, analysis is essential.



Learning language, indexing information, and building knowledge take time.



Precocious Child

GPT models learn language and relationships through training on huge amounts of foundational data.



Librarian & Card Catalog

A vector-based RAG solution can quickly put a wealth of new information in your hands.



Research Assistant

GraphRAG studies all available information to partner with you on your knowledge journey.

> I need to author a report on considerations for extra-solar exploration.



Precocious Child (LLMs)

- Shares facts that it knows about space, potentially missing the topic entirely.
- May tend to repeat itself.
- Lacks the ability to think critically about the query.



Librarian & Card Catalog (RAG)

- Will be able to quickly help you find information relevant to astronomy.
- May not be able to suggest related fields (e.g., ethics, finances, environment)
- Most analysis still required of the user.



Research Assistant

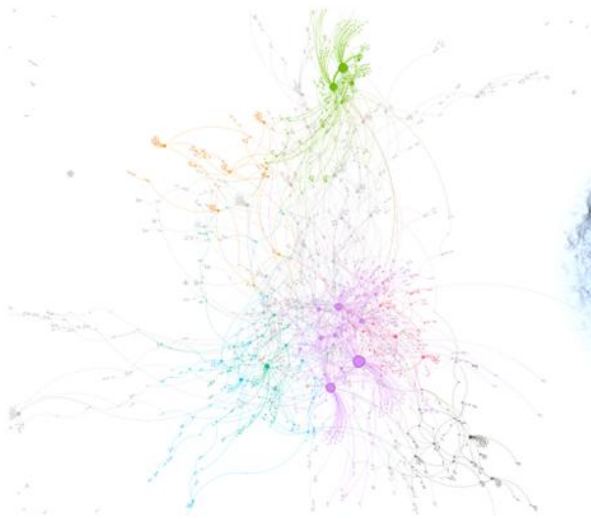
- Familiar with available data, with perspective tailored by domain expertise.
- Able to interrogate broad view of understanding as well as recall specific details.
- Partners with user on analysis journey.

Turning inward for inspiration, GraphRAG emulates the human brain.

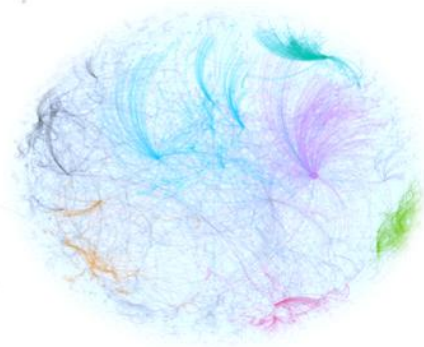
- Highly-connected web of ideas.
- Topics related by clusters of relations, not semantic similarity.
- Organized progressively to work with fuzzy language.
- Abstract new ideas to our existing mental models.
- Influenced by domain expertise.



Organizing Knowledge



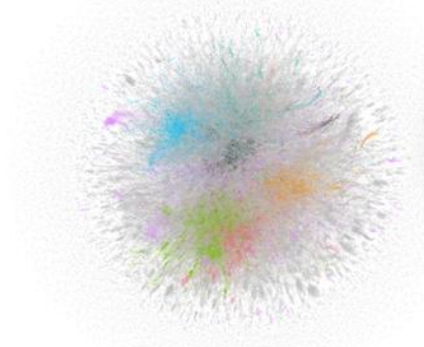
My writings



A prominent Scholar



A niche



A field of
study



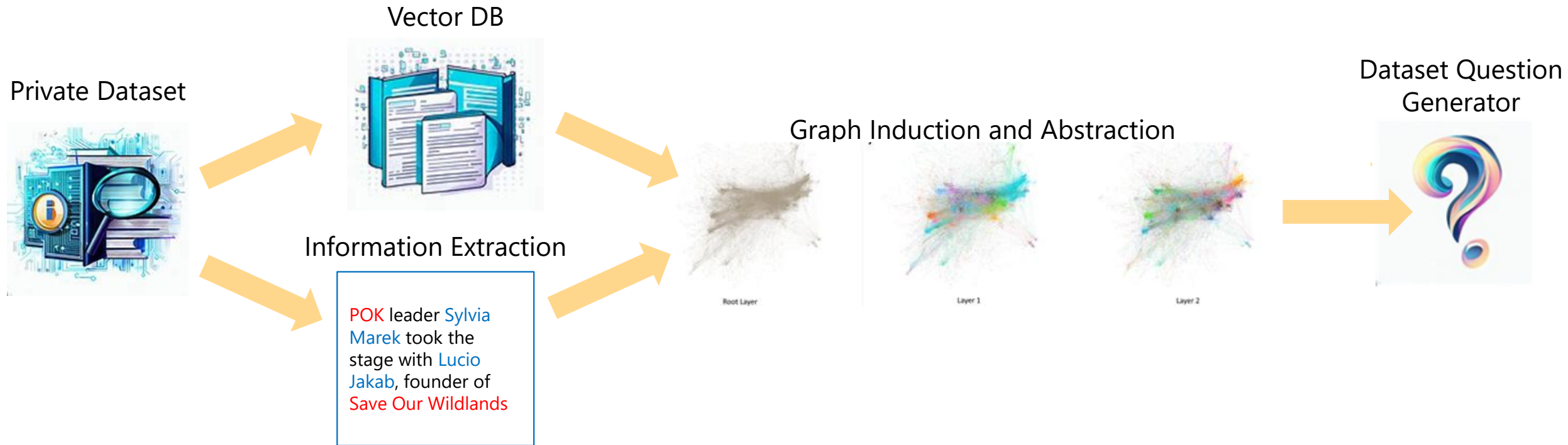
An academic
discipline

Preprocessing Pipeline – Traditional RAG



- ✓ Information lookup that is too big for context window
- ✗ Questions that require multi-hop reasoning
- ✗ Topics that have thin connections across many documents
- ✗ Abstractions and thematic retrievals

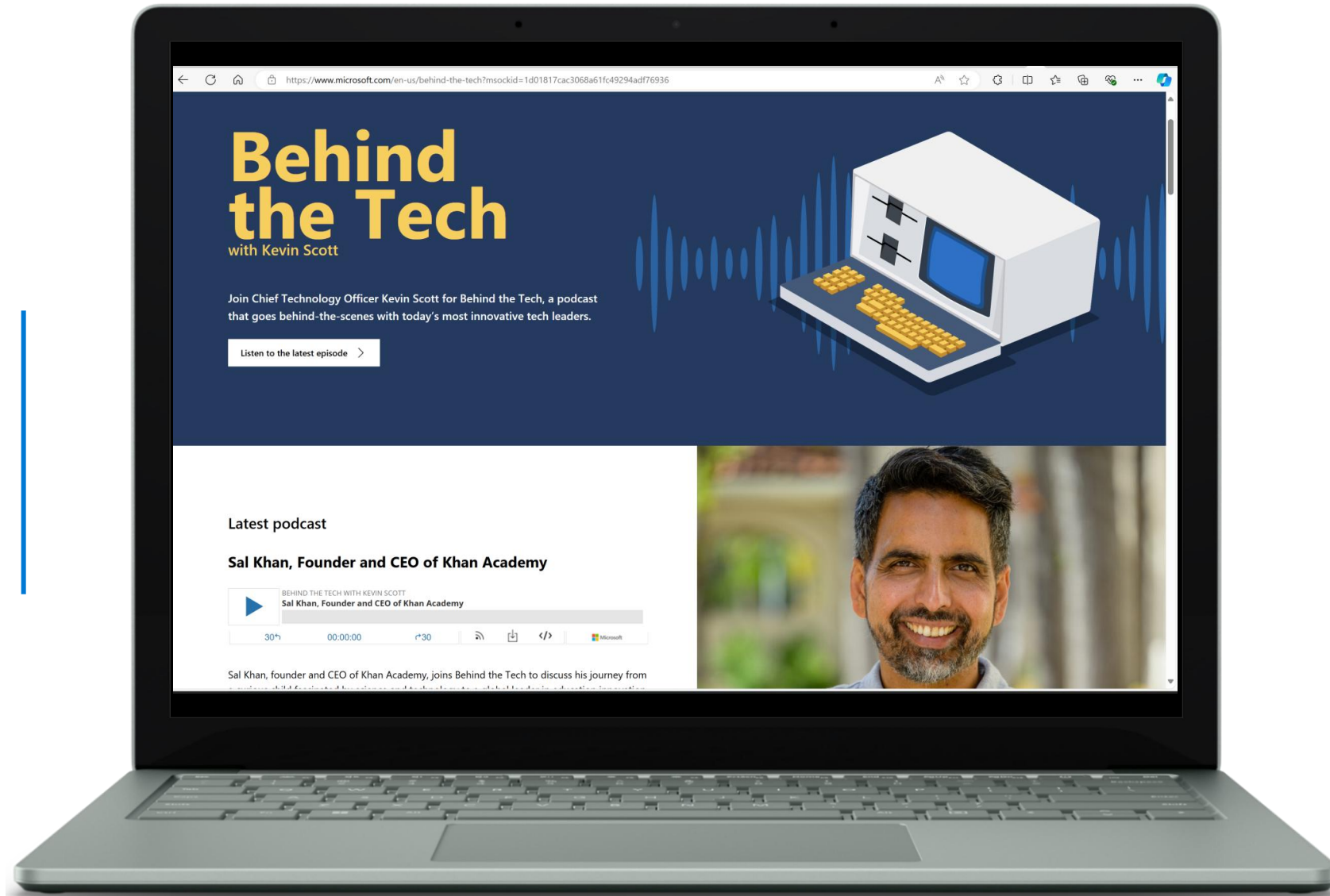
Preprocessing Pipeline – GraphRAG



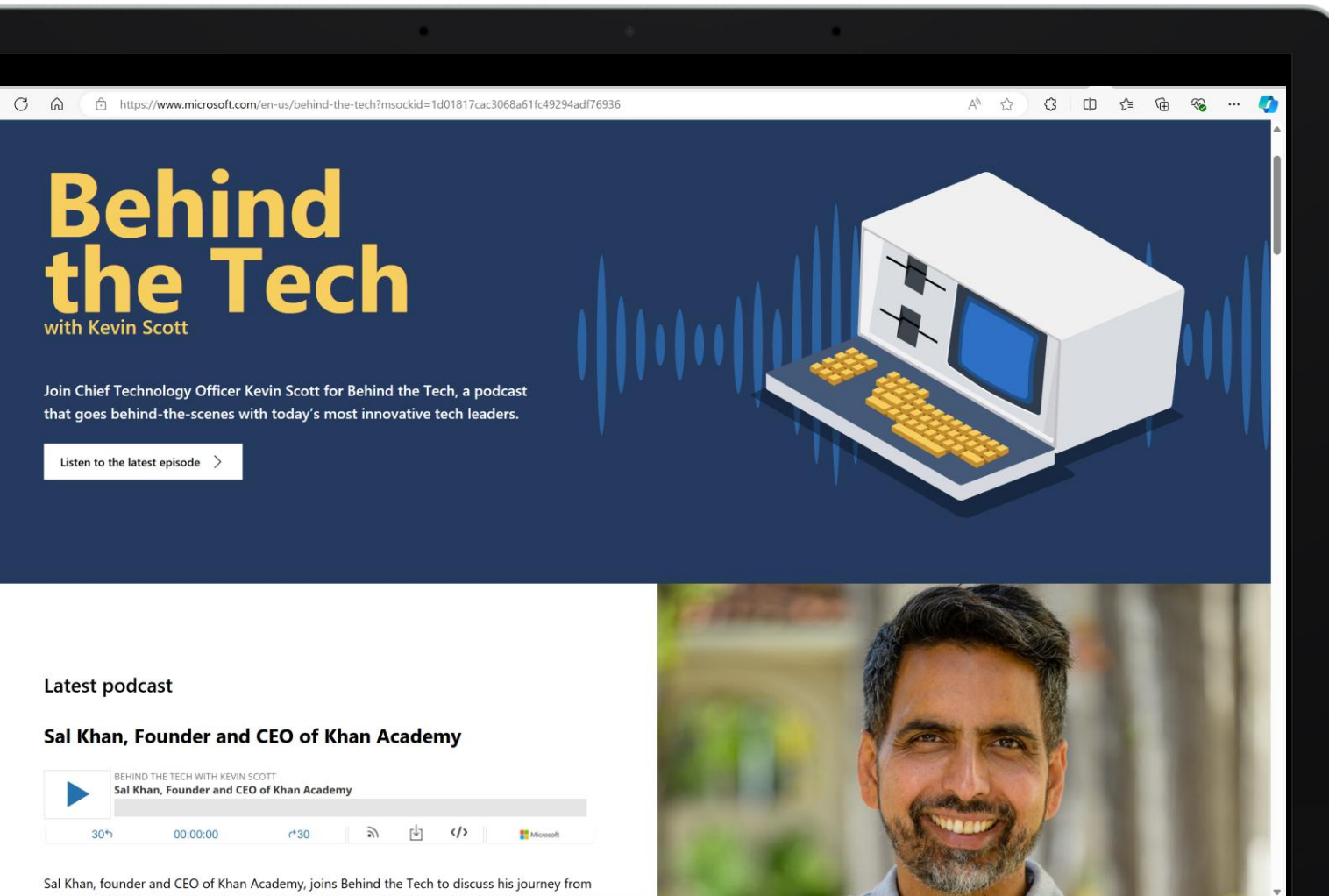
- ✓ Information lookup that is too big for context window
- ✓ Questions that require multi-hop reasoning
- ✓ Topics that have thin connections across many documents
- ✓ Abstractions and thematic retrievals

EXAMPLE

Behind the Tech



I need to know what my CTO and executives know and think about Bio...



Chatbot Interface

USER

Give me a top-level summary of all biology discussions in Kevin Scott's Behind the Tech podcast.

ASSISTANT

As of my last update in October 2023, I don't have access to specific episode details or summaries of Kevin Scott's "Behind the Tech" podcast. However, I can provide a general idea of what you might expect from biology-related discussions on a tech-focused podcast like this one.

1. **Biotechnology Innovations**: Discussions might cover the latest advancements in biotechnology, including CRISPR, gene editing, and synthetic biology. These topics often explore how technology is revolutionizing biological research and applications.
2. **Health Tech**: Episodes could delve into how technology is transforming healthcare, from telemedicine to wearable health devices, and the integration of AI in diagnostics and treatment plans.
3. **Bioinformatics**: The intersection of biology and data science might be a key topic, focusing on how big data and computational tools are used to understand complex biological systems and diseases.
4. **Environmental Tech**: Conversations might include how technology is being used to address environmental challenges, such as climate change, conservation, and sustainable agriculture.
5. **Interviews with Experts**: The podcast might feature interviews with leading biologists, bioengineers, and tech entrepreneurs who are at the forefront of integrating technology with biological sciences.

For specific details, I recommend checking the episode list and summaries on the official "Behind the Tech" podcast website or your preferred podcast platform.

Answer context: Fixed number of text chunks of raw documents

1,076 tokens used, 1 LLM calls, 18 seconds elapsed.

Summary of Biology Discussions in the Podcast

Introduction to the Podcast

The podcast episode features Dr. Eric Horvitz, Chief Scientific Officer at Microsoft, who delves into various topics, including the intersection of biology, artificial intelligence (AI), and high-performance computing. The conversation is described as fascinating and covers a broad range of subjects relevant to the year's themes [Data: Sources (0)].

Early Influences and Education

Dr. Horvitz shares his early experiences with science and education, highlighting the significant impact of his teachers. He recalls being elected as the chairperson of the science club in third grade and engaging in projects related to wind speed and solar energy. These early experiences, driven by curiosity and books, laid the foundation for his later interest in science and technology [Data: Sources (0)].

Inspirational Educators

Dr. Horvitz emphasizes the role of inspirational educators in shaping his scientific journey. He mentions several teachers by name, such as Mrs. Frank, Mrs. O'Hara, and Mr. Wilmott, who encouraged his interests and celebrated his achievements, such as winning a science fair in sixth grade. These educators acted as pivotal figures, guiding him towards a career in science [Data: Sources (0)].

Intersection of Biology, AI, and High-Performance Computing

The podcast touches on the intersection of biology with AI and high-performance computing, which has been a significant theme for the year. Although specific details of these discussions are not provided in the text, it is clear that Dr. Horvitz's work involves integrating these fields to advance scientific understanding and technological innovation [Data: Sources (0)].

Conclusion

The episode underscores the importance of curiosity, mentorship, and interdisciplinary approaches in scientific endeavors. Dr. Horvitz's reflections on his educational journey and the role of influential teachers provide valuable insights into the factors that contribute to a successful career in science and technology. Overall, the podcast offers a rich exploration of how early educational experiences and the integration of diverse scientific fields can drive innovation and discovery in biology and beyond.

Answer context: AI-generated network reports covering all input documents

610,968 tokens used, 38 LLM calls, 38 seconds elapsed.

Summary of Biology Discussions in the "Behind the Tech" Podcast

Introduction to the Podcast

The "Behind the Tech" podcast, hosted by Kevin Scott, covers a wide array of topics within the field of biology, often intersecting with technology, healthcare, and scientific research. Below is a comprehensive summary of the key biology discussions featured in the podcast.

Impact of COVID-19

The COVID-19 pandemic is a recurring theme, significantly influencing discussions on biomedicine, biotechnology, and public health. The pandemic has driven advancements in vaccine development, particularly mRNA vaccines, and has highlighted the importance of protein structures like the SARS-CoV-2 spike protein in therapeutic design [Data: Reports (698, 697, 822, 908, +more)]. The role of scientific research in addressing global health crises is emphasized, with notable contributions from figures like Dr. David Baker and Neil King in developing coronavirus vaccines [Data: Reports (697, 709, 651, +more)].

Protein Design and Synthetic Biology

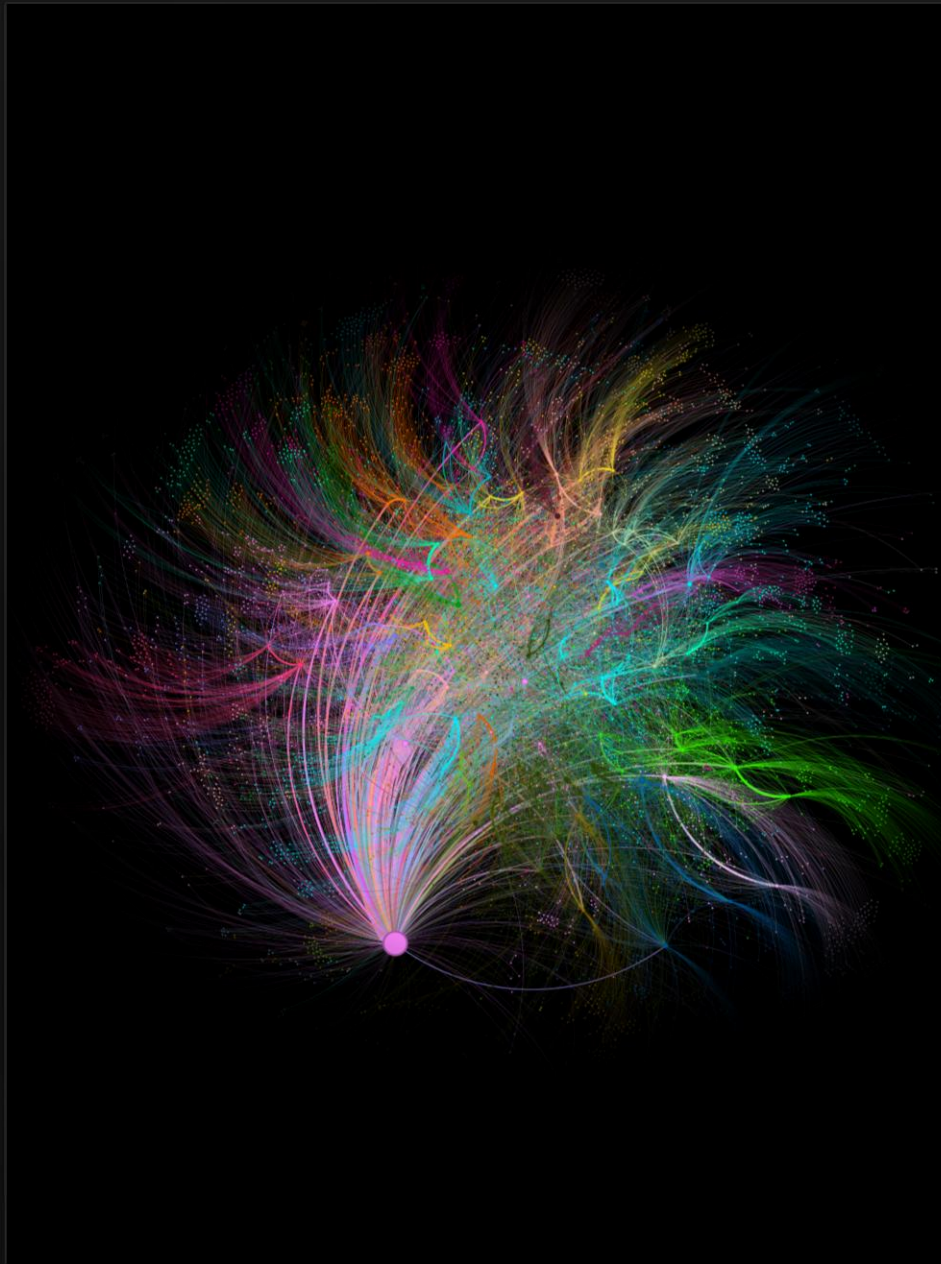
Protein design and synthetic biology are central topics, with significant contributions from experts like Dr. David Baker and Drew Endy. Discussions cover the design of therapeutic proteins, the development of next-generation vaccines, and the potential of synthetic biology to revolutionize healthcare and biotechnology [Data: Reports (697, 876, 709, 707, +more)]. The Institute for Protein Design at the University of Washington, directed by David Baker, is frequently mentioned for its pioneering work in protein structure prediction and therapeutic design [Data: Reports (709, 706, 707, +more)].

Genetic Engineering and Computational Biology

Genetic engineering is highlighted for its role in vaccine development and bioengineering advancements. The podcast delves into the applications of CRISPR and other gene editing technologies in neuroscience and medicine [Data: Reports (648, 491, 901, +more)]. Computational biology is another key area, integrating computer science and biology to model biological systems and design new proteins. This interdisciplinary approach is crucial for modern biological research and innovation [Data: Reports (707, 646, 703, +more)].

Neuroscience and AI

Neuroscience discussions often intersect with AI and machine learning, exploring how these technologies can enhance our understanding of the brain and improve diagnostics and treatments for neurological disorders. Notable figures like Eric Horvitz and Fei-Fei Li contribute to these discussions, emphasizing the potential of AI



Answer context: AI-generated network reports covering all input documents
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Large Context Windows and GraphRAG

GraphRAG enables large dataset reasoning on private data

- e.g., arXiv paper analysis tests have run > **50M tokens** for even sub-topical explorations

We evaluate GraphRAG against Large Context Windows

- “Behind the Tech” podcast transcripts represented as ~900K token dataset
- Fits within Gemini’s 1M token context window

Typical queries run faster even with more depth:

- Gemini 1.5: ~90 seconds
- GraphRAG: ~50 seconds

GraphRAG benefits from larger context windows, too

- More detail is added to returned corpus
- More capacity is available for tangential connections

GraphRAG's emergent scalability

- GraphRAG is the subject of ongoing research, and its upper limits continue to be explored.
- GraphRAG is **most effective** with corpora of **>100K tokens**.
- GraphRAG's been evaluated against corpora up to **230K pages of raw text (650MB, 160M tokens)**
- GraphRAG's largest knowledge graph extracted contains **900K nodes and 2.6M edges**.

Comparing approaches

	No Augmentation	RAG	GraphRAG
Context Awareness	None	Some	Deep
“Direct hit” search results	✗	✓	✓
Topically relevant connections	✗	✗	✓
Deep and sparse connections	✗	✗	✓
“Question behind the question”	✗	✗	partial
Hallucination Risk	High	Low	Very Low
Supports Hallucination Checking	✗	✓	✓
Provides source references	✗	✓	✓
Answers direct questions	✗	Usually	✓
Answers nuanced questions	✗	✗	✓
Explores adjacent topics and multiple perspectives	✗	✗	partial
Reasons across many documents	✗	~10	✓
Time to response	very fast (~5s)	fast (~5-10s)	medium (~10-30s)

GraphRAG Open-Source Program

Python Library

github.com/microsoft/graphrag

- Installable via pip
- provides both an indexing engine and a query engine
 - graph database is not necessary
- can only process CSV or TXT files
- saves off the knowledge graph to disk as .parquet files
- querying – 2 types of queries
 - global search – great for broad queries about an entire dataset
 - local search – great for narrow-focused queries
 - DRIFT search – still experimental; in between global/local

Solution Accelerator [public archive]

github.com/Azure-Samples/graphrag-accelerator

- Azure Reference Implementation for GraphRAG
- Includes many security best-practices but is still not intended for production use.
- Deployed via single line bash script
- graphrag library deployed to AKS, fronted by APIM
- Hardcoded AOAI TPM/RPM settings
- Optional 'Developer Mode' -g
- Example Jupyter notebooks for quickstart; optional front-end streamlit app
- Fine tuning – uses an optional "Auto-template generation" capability to improve quality of knowledge graph creation; can be adjusted by hand as well.

FAQ

Why use Microsoft's GraphRAG instead of [other open-source project]?

- Microsoft's solution is the most complete end-to-end solution for solving this problem and you can expect continued innovation because *we use it ourselves*.

How does this work with knowledge already memorized by the model?

- Knowledge memory is fragile and most useful for working through difficult semantics. Having good background information in the model only helps make GraphRAG more powerful in deep niche topics.

What's the easiest way to try this?

- If you are a dev, *start with the python library*. Once tested, the Azure Accelerator will get you to a stable API oriented solution quickly.

When should I use 'regular' RAG and not bother with GraphRAG?

- If your problem is a "dictionary lookup" where there are clear keywords to search on.

FAQ

How will this stay relevant as AI models advance?

- GraphRAG is designed to organize knowledge like people do, so as the AI models advance (cheaper to run, bigger contexts, higher quality), we expect GraphRAG to improve in value alongside them.

How do I control costs?

- GraphRAG can be relatively expensive to run as it does a lot of thinking upfront. The best way to control cost is to use GraphRAG on data where you will have high value and use vector RAG for lower-value datasets that are just additional references.

How do I personalize it to my domain or way of thinking?

- GraphRAG offers customization options for the indexing and query steps, you can create custom types or modify the query mechanism to have a persona that aligns to your usecase.

Section Summary

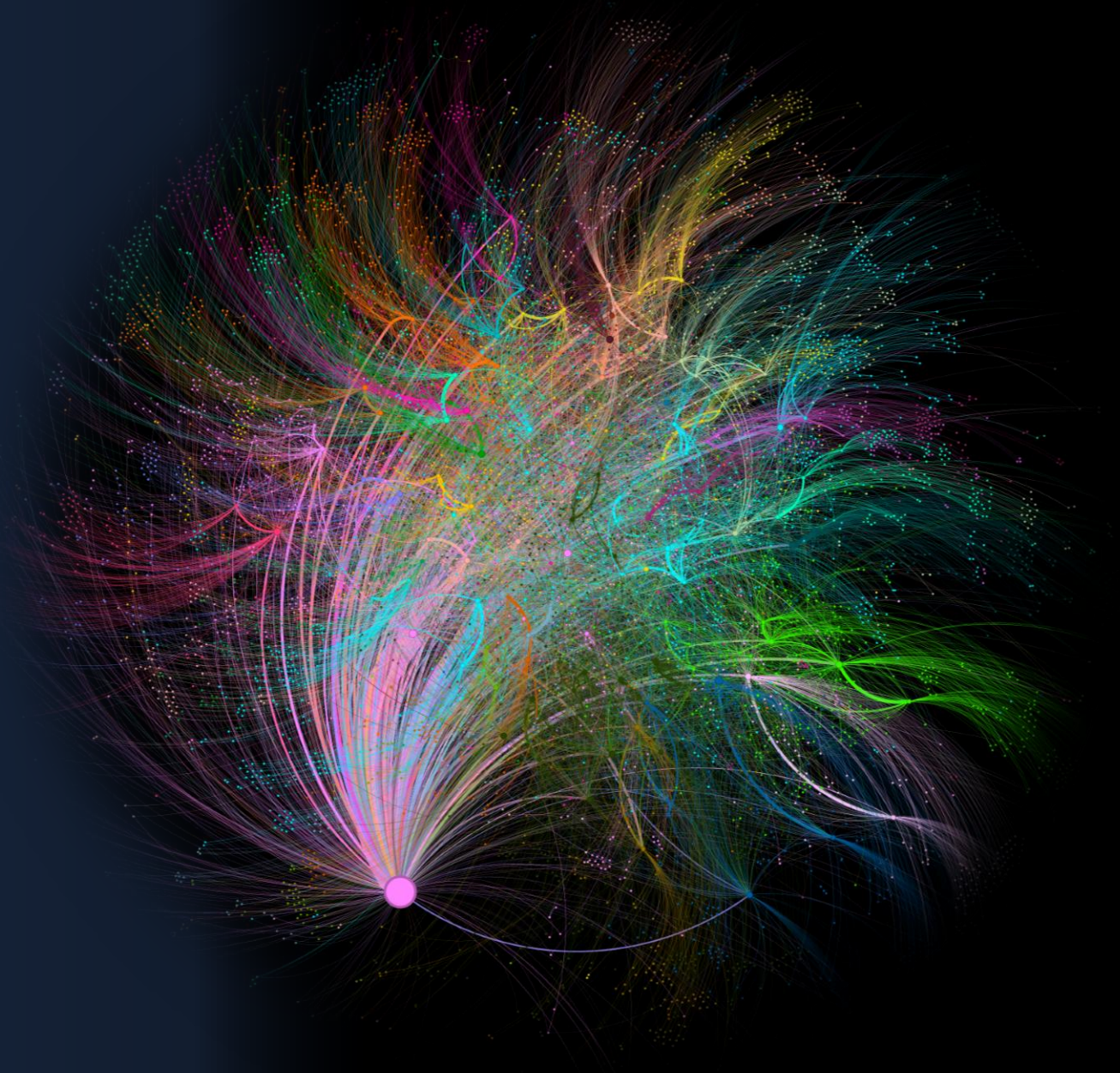
GraphRAG is a novel approach to RAG that uses LLM-generated Knowledge Graphs to reason about complex, noisy, and deep data sets.

GraphRAG is designed with a simple premise: to emulate how the human brain organizes, relates, and retrieves information.

GraphRAG has showed improvement in quality and speed when compared with LLMs with large context windows.

GraphRAG can be costlier due to upfront processing. To control costs, use GraphRAG for high-value data and vector RAG for lower-value, supplementary datasets.

GraphRAG is available as both a Python Library and an Azure Solution Accelerator.



Getting started

With the GraphRAG Python Library

Prerequisites



**A Python 3.10 to
3.12 environment**



**An Azure Open AI
Service resource**



**A text embedding
model (ada-002)**



**An LLM
(GPT-4o-mini)**

Getting Started

Create working dir

```
mkdir graphrag && cd graphrag
```

Create and activate venv

```
python3 -m venv venv
```

```
cd venv && source bin/activate
```

Install GraphRAG

```
pip install graphrag
```

Initialize workspace named ragtest

```
mkdir -p ragtest/input
```

```
graphrag init --root ragtest
```

Installation & Setup

Create a working directory

Create and activate a Python virtual environment

Install GraphRAG from **pypi**

Initialize a workspace
(**ragtest**)

```
# Show all files in 'ragtest'
find -f ./ragtest
  settings.yaml
  .env
  prompts
  ...
  prompts/summarize_descriptions.txt
  prompts/community_report.txt
  prompts/entity_extraction.txt
```

Explore freshly initialized workspace

GraphRAG `init` will create a `settings.yaml` and a `.env` file that we need to update in the next step.

It will also generate several default prompts that are used for data indexing and query.

```
# Download A Christmas Carol
```

```
curl -L https://aka.ms/dickens/xmas \  
-o ./ragtest/input/book.txt
```

```
# Verify download
```

```
wc ./ragtest/input/book.txt
```

```
#      3972      32457     189225 book.txt
```

Get Source Data

For this first index and query, we will utilize "A Christmas Carol" by Charles Dickens from Project Gutenberg.

```
# Add AOAI API KEY to .env
vi ragtest/.env

GRAPHRAG_API_KEY=<API_KEY>
~
~
```

Update **.env** with AOAI API Key

The API Key can be obtained from the AOAI Service Portal within Azure AI Foundry (ai.azure.com).

Getting Started

```
# Add connection info to settings.yaml
vi ragtest/settings.yaml
```

```
...
llm:
  ...
  type: azure_openai_chat
  model: gpt-4o-mini
  api_base: https://<instance>.openai.azure.com
  api_version: 2024-02-15-preview
  deployment_name: gpt-4o-mini

...
embeddings:
  ...
  llm:
    ...
    type: azure_openai_chat
    model: text-embedding-ada-002
    api_base: https://<instance>.openai.azure.com
    api_version: 2024-02-15-preview
    deployment_name: text-embedding-ada-002
```

~

Update **settings.yaml** with AOAI and Model connection information

This can all be obtained from
the AOAI Service Portal
within Azure AI Foundry
(**ai.azure.com**).

Be sure to update both the
llm and **embeddings** blocks.

```
# Add connection info to settings.yaml
vi ragtest/settings.yaml

...
snapshots:
  graphml: true
```

Update `settings.yaml` to create graphml snapshots

This is an optional configuration that tells GraphRAG to output a graphml file that can be later used for graph visualization.

My first index

This process will run the indexing pipeline. It will take some time, as the LLM is analyzing *A Christmas Carol* and creating the knowledge graph.

When complete, we can review what it did inside of the output folder.

```
# Run GraphRAG Indexing Pipeline
```

```
graphrag index --root ./ragtest
```



All workflows completed successfully.

```
# Review output directory
```

```
ls ./ragtest/output
```

```
...
```

```
create_final_communities.parquet  
create_final_community_reports.parquet  
create_final_documents.parquet  
create_final_entities.parquet  
create_final_nodes.parquet  
create_final_relationships.parquet  
create_final_text_units.parquet  
graph.graphml  
lancedb
```

Overview of a GraphRAG Index



What is a GraphRAG index?

A collection of tables and embeddings that describe the structure of a knowledge graph.



How can we create a new index?

The GraphRAG library offers an indexer client that allows us to build new indexes from a text corpus with no upfront development.



What file types are supported?

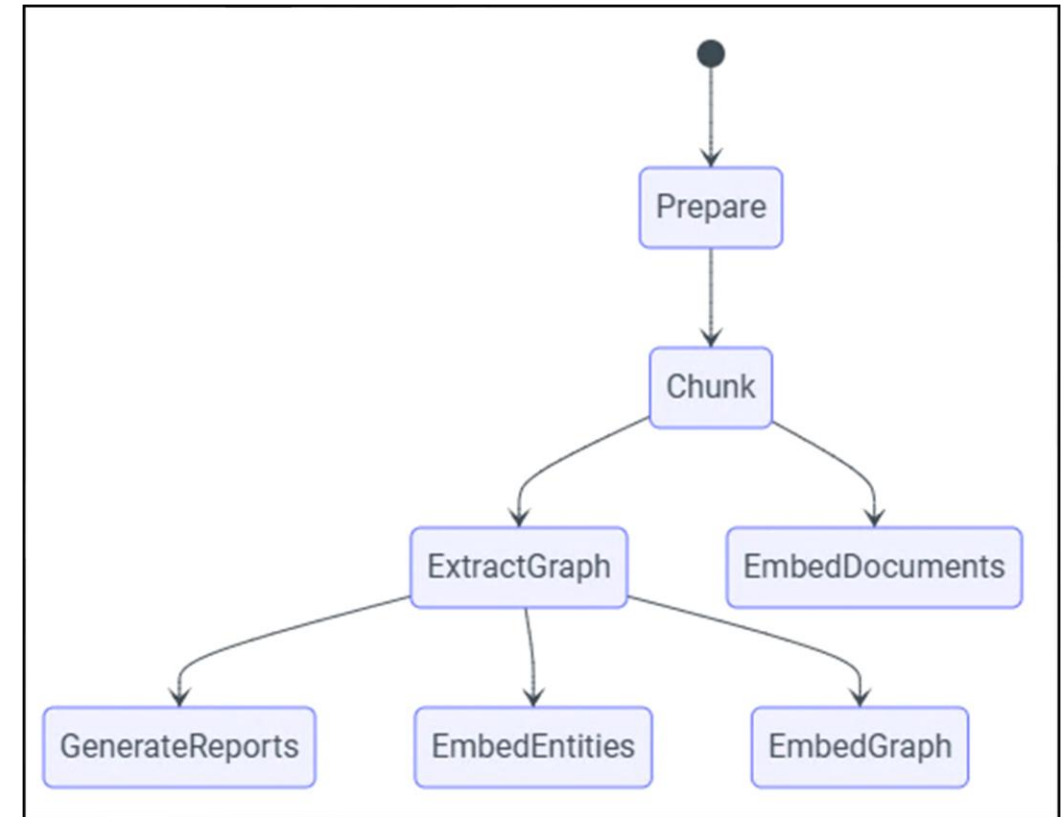
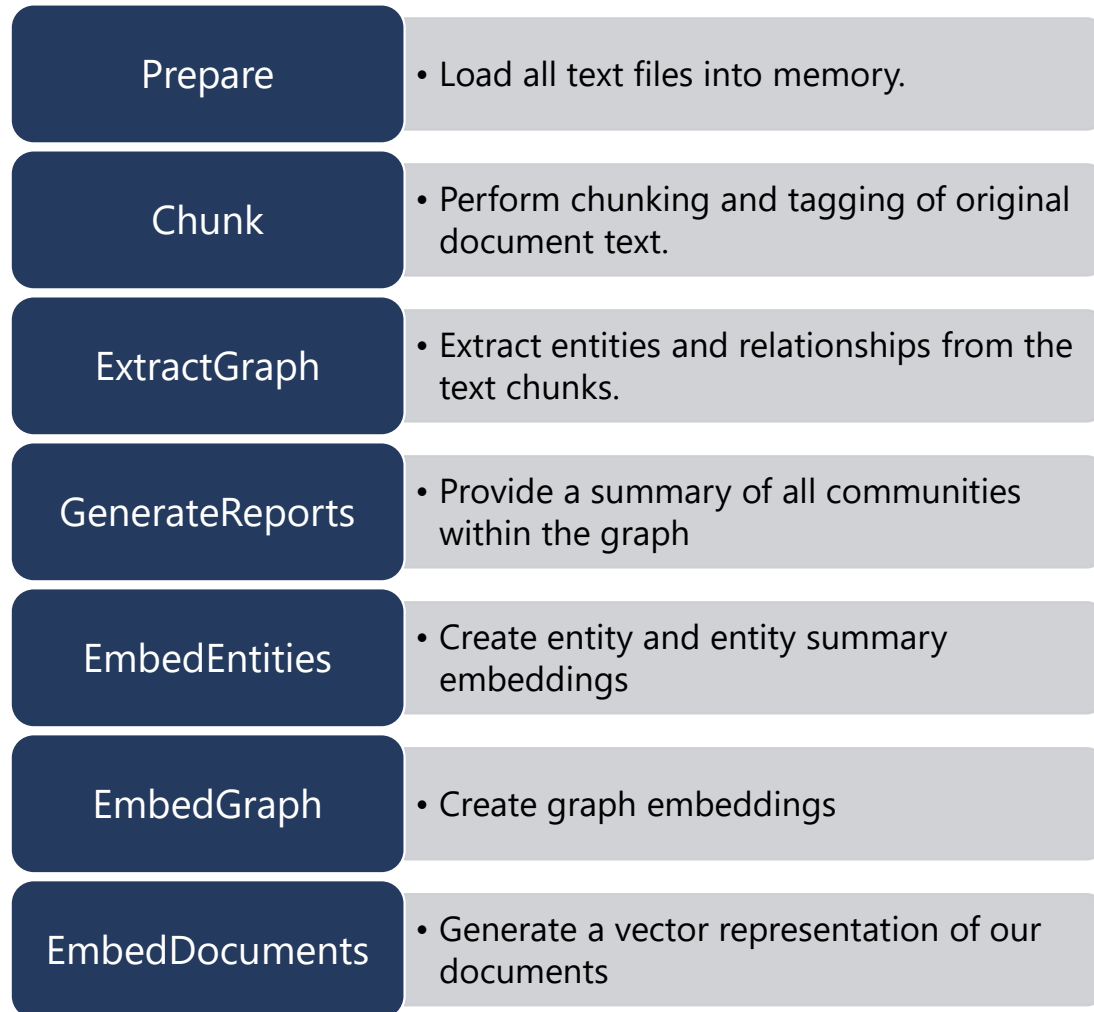
The GraphRAG indexer library only supports text files with a **".txt"** extension.



How can I influence how GraphRAG indexes?

GraphRAG's indexer library supports both auto-tuning and manual-tuning through configuration. We'll explore this later in the workshop.

The GraphRAG Indexing Workflow



Querying - Global

```
# Execute a global query
graphrag query \
--root ./ragtest \
--method global \
--query "What are the top themes in this story?"
```

SUCCESS: Global Search Response:

The story presents several prominent themes that intertwine to convey its central messages. Below are the key themes identified:

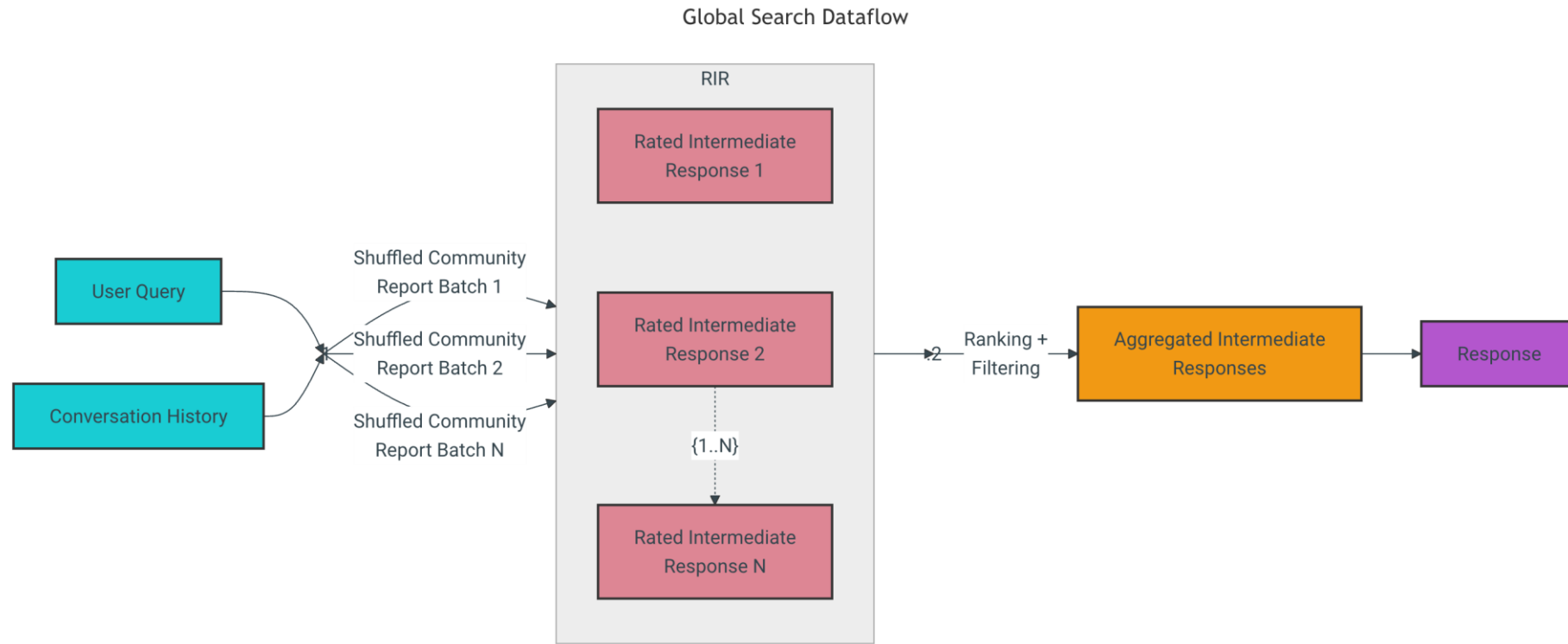
Transformation and Redemption

The theme of transformation is pivotal, particularly illustrated through the character of Ebenezer Scrooge. His evolution from a miser to a generous benefactor is catalyzed by visits from the Three Spirits, each representing different aspects of his life and potential future. This transformation underscores the potential for change and the importance of compassion in human

My first queries

First, we will conduct a **Global Query**.

The global search technique analyzes all AI-generated community reports through a map-reduce method. Although resource-intensive, it provides valuable insights for questions that require a **thorough, broad understanding** of the dataset, like themes or concepts.



Global Query utilizes a **map-reduce** approach

- Segments Community Reports into text chunks
- Generates and rates intermediate responses.
- Aggregates most relevant responses and uses LLM to generate final response.

Execute a local query

```
graphrag query \  
--root ./ragtest \  
--method local \  
--query "Who is Scrooge and what are his  
main relationships?"
```

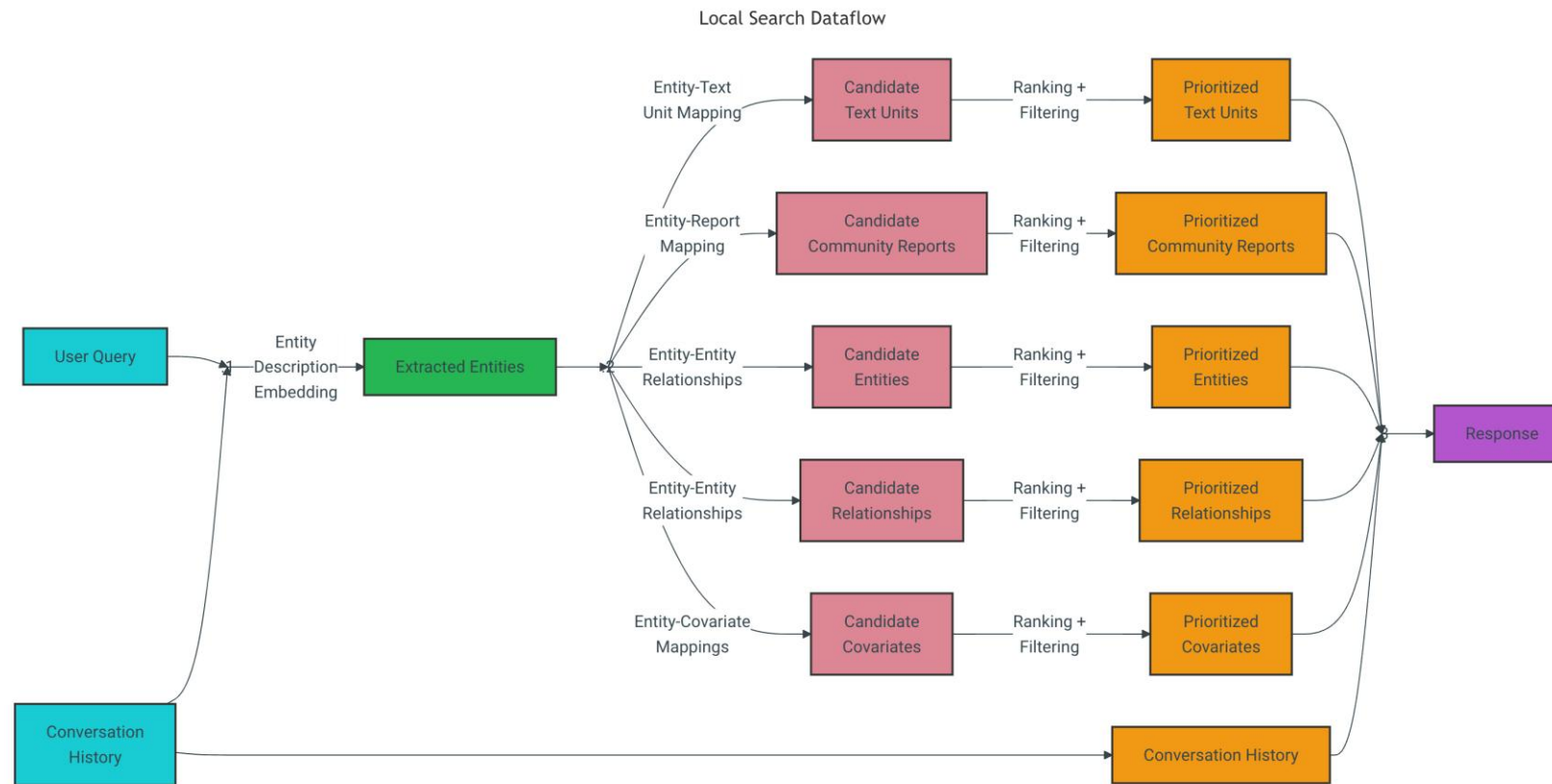
SUCCESS: Local Search Response:
Who is Scrooge?

Ebenezer Scrooge is the central character in Charles Dickens' classic novella "A Christmas Carol." Initially depicted as a miserly old man, Scrooge embodies greed, selfishness, and a disdain for Christmas and human connection. His character is marked by a cold demeanor and a lack of empathy, particularly towards the poor and those in need. Scrooge's transformation begins when he is visited by the ghost of his former business partner, Jacob Marley.

My first queries

Now, we will conduct a **Local Query**.

The local search method combines structured data from the knowledge graph with unstructured data from the input documents to augment the LLM context with relevant entity information at query time. It is **well-suited for answering questions** that require an understanding of specific entities mentioned in the input documents.



Local Query uses entity-based reasoning to find information

- Extracts entities in user query and looks them up in knowledge graph.
- Prioritizes and filters results.
- Aggregates most relevant knowledge graph nodes and edges and uses LLM to generate final response.

Querying - DRIFT

```
# Execute a DRIFT query
graphrag query \
--root ./ragtest \
--method drift \
--query "Who is Scrooge and what are his
main relationships?"
```

SUCCESS: DRIFT Search Response:
Overview of Ebenezer Scrooge and His Key Relationships

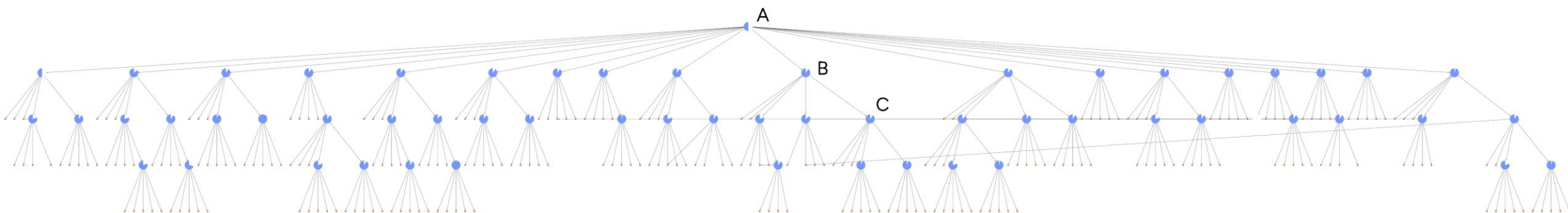
Ebenezer Scrooge is the main character of Charles Dickens' novella "A Christmas Carol," renowned for his transformation from a miserly, cold-hearted individual to a compassionate figure embodying the spirit of Christmas. This metamorphosis is influenced significantly by key relationships in his life, which highlight themes of redemption, charity, and the

My first queries

DRIFT Search was recently announced in October 2024.

(Dynamic Reasoning and Inference with Flexible Traversal)

A new approach to local queries that includes 'global' information in a new iterative search process.



Drift Queries expand the starting point and ask follow-ups.

- Integrates community info for diverse facts.
- Combines Global and Local methods to balance cost and quality.
- Uses adaptive query refinement and ranks answers by relevance.
- Suitable for a wide range of queries.
- Still experimental.

Section Summary

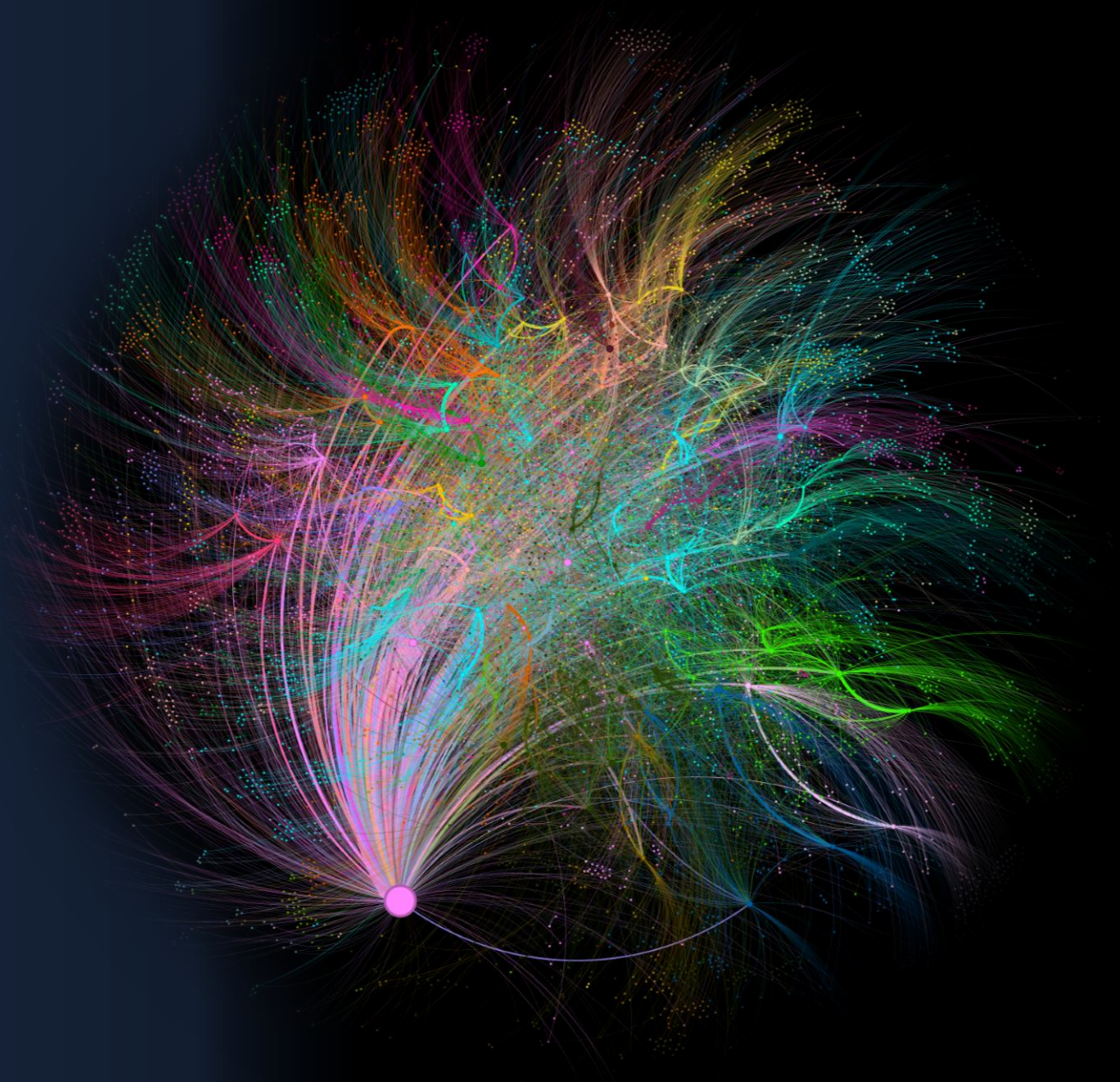
Getting started with the GraphRAG Library requires a Python 3.10 to 3.12 environment and access to an Azure OpenAI Service resource.

The GraphRAG library is available on Pypi and can be installed with `pip install graphrag`.

The GraphRAG Indexer uses LLMs to extract and structure data from unstructured text through customizable workflows, embedding entities, relationships, and claims into graph and textual vector spaces.

GraphRAG's **Global Search** uses community summaries and a map-reduce approach to answer questions that require a **thorough, broad understanding of a dataset**.

GraphRAG's **Local Search** uses entity-based reasoning to quickly answer questions that require an **understanding of specific entities mentioned in the input documents**



Introducing Domain Expertise

Through prompt engineering

GraphRAG uses Prompt Tuning to emulate domain expertise.

Auto Tuning

- Automatically generate indexing prompts based on your data corpus.
- No upfront code, **turnkey solution**.
- Works well in most domains.
- We strongly recommend using Auto Tuning when your corpus **spans multiple domains**.
- Might not be the best solution for niche corpuses.

Manual Tuning

- For advanced users.
- Requires knowledge of the underlying data.
- Best for datasets spanning **a single domain**.
- Requires oversight and testing.

```
# After having previously run  
graphrag init
```

```
# Run GraphRAG auto tuning process
```

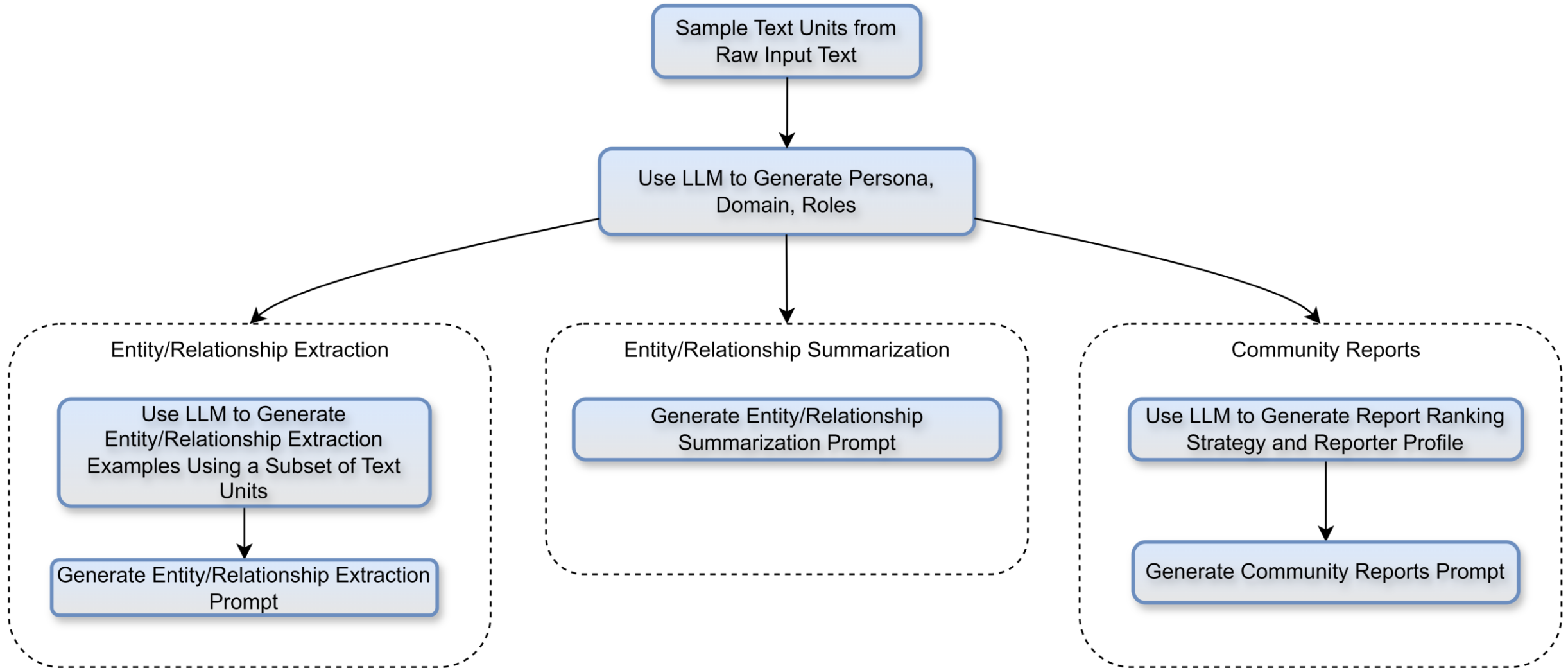
```
graphrag prompt-tune \  
  --root ./ragtest \  
  --config ./ragtest/settings.yaml \  
  --output ./ragtest/prompts \  
  --domain "Literary Analyst"
```

Discovering Domain Expertise

The Auto Tuning process will take a sampling of data from your corpus to devise a relevant domain, system personas, and entity types.

It will then apply this information to some sample data and generate system prompts for use by the Indexing Process.

Auto Tuning



graphrag prompt-tune command line options

Option	Description	Required?	Default
--config	The path to the configuration file. This is required to load the data and model settings.	Required	
--root	The data project root directory, including the config files (YML, JSON, or .env).	Optional	Current directory
--domain	The domain related to your input data (e.g., 'space science', 'microbiology').	Optional	Inferred from data
--method	The method to select documents. Options: all, random, auto, top.	Optional	Random
--limit	The limit of text units to load when using random or top selection.	Optional	15
--language	The language to use for input processing. If different from inputs' language, LLM will translate.	Optional	Auto-detected
--max-tokens	Maximum token count for prompt generation.	Optional	2000
--chunk-size	The size in tokens to use for generating text units from input documents.	Optional	200
--n-subset-max	The number of text chunks to embed when using auto selection method.	Optional	300
--k	The number of documents to select when using auto selection method.	Optional	15
--min-examples-required	The minimum number of examples required for entity extraction prompts.	Optional	2
--no-entity-types	Allow the LLM to discover and extract entities automatically. Recommended for diverse data.	Optional	
--output	The folder to save the generated prompts.	Optional	"prompts"

Comparing the Defaults with Auto Tuned Prompts

Default `entity_extraction.txt`

-Goal-

Given a text document that is potentially relevant to this activity and a list of entity types, identify all entities of those types from the text and all relationships among the identified entities.

-Steps-

1. Identify all entities. For each identified entity, extract the following information:

- entity_name: Name of the entity, capitalized
- entity_type: One of the following types: [{entity_types}]
- entity_description: Comprehensive description of the entity's attributes and activities

Format each entity as

("entity"{tuple_delimiter}<entity_name>{tuple_delimiter}<entity_type>{tuple_delimiter}<entity_description>)

...

#####

-Examples-

#####

Example 1:

Entity_types: ORGANIZATION,PERSON

Text:

The Verdantis's Central Institution is scheduled to meet on Monday and Thursday, with the institution planning to release its latest policy decision on Thursday at 1:30 p.m. PDT, followed by a press conference where Central Institution Chair Martin Smith will take questions. Investors expect the Market Strategy Committee to hold its benchmark interest rate steady in a range of 3.5%-3.75%.

#####

Output:

("entity"{tuple_delimiter}CENTRAL
INSTITUTION{tuple_delimiter}ORGANIZATION{tuple_delimiter}The Central Institution is the
Federal Reserve of Verdantis, which is setting interest rates on Monday and Thursday)

{record_delimiter}

("entity"{tuple_delimiter}PERSON{tuple_delimiter}MARTIN SMITH{tuple_delimiter}Chair of the Central Institution

Tuned `entity_extraction.txt`

-Goal-

Given a text document that is potentially relevant to this activity and a list of entity types, identify all entities of those types from the text and all relationships among the identified entities.

-Steps-

1. Identify all entities. For each identified entity, extract the following information:

- entity_name: Name of the entity, capitalized
- entity_type: One of the following types: [person, spirit, family, holiday, emotion, event, character, relationship]
- entity_description: Comprehensive description of the entity's attributes and activities

Format each entity as

("entity"{tuple_delimiter}<entity_name>{tuple_delimiter}<entity_type>{tuple_delimiter}<entity_description>)

...

#####

-Examples-

#####

Example 1:

entity_types: [person, spirit, family, holiday, emotion, event, character, relationship]

text:

I shouldn't be at all surprised--mark what I say!--if he got

Peter a better situation.'

'Only hear that, Peter,' said Mrs. Cratchit.

'And then,' cried one of the girls, 'Peter will be keeping company with some one, and setting up for himself.'

'Get along with you!' retorted Peter, grinning.

'It's just as likely as not,' said Bob, 'one of these days; though there's plenty of time for that, my dear. But, however and whenever we part from one another, I am sure we shall none of us forget poor Tiny Tim--shall we--or this first parting that there was among us?'

Comparing the Defaults with Auto Tuned Prompts

Default `community_report.txt`

You are an AI assistant that helps a human analyst to perform general information discovery. Information discovery is the process of identifying and assessing relevant information associated with certain entities (e.g., organizations and individuals) within a network.

Goal

Write a comprehensive report of a community, given a list of entities that belong to the community as well as their relationships and optional associated claims. The report will be used to inform decision-makers about information associated with the community and their potential impact. The content of this report includes an overview of the community's key entities, their legal compliance, technical capabilities, reputation, and noteworthy claims.

Report Structure

The report should include the following sections:

- TITLE: community's name that represents its key entities - title should be short but specific. When possible, include representative named entities in the title.
- SUMMARY: An executive summary of the community's overall structure, how its entities are related to each other, and significant information associated with its entities.
- IMPACT SEVERITY RATING: a float score between 0-10 that represents the severity of IMPACT posed by entities within the community. IMPACT is the scored importance of a community.
- RATING EXPLANATION: Give a single sentence explanation of the IMPACT severity rating.
- DETAILED FINDINGS: A list of 5-10 key insights about the community. Each insight should have a short summary followed by multiple paragraphs of explanatory text grounded according to the grounding rules below. Be comprehensive.

Tuned `community_report.txt`

You are an expert `Literary Analyst`. You are skilled at dissecting texts to uncover `underlying themes, character dynamics, and narrative structures`. You are adept at helping people understand the intricate relationships and frameworks within `literary communities`, facilitating deeper insights into how these elements influence interpretation and appreciation of literature.

Goal

Write a comprehensive assessment report of a community taking on the `role of a **Role: Literary Community Analyst**`

A `literary community analyst` tasked with examining the text from Charles Dickens' "`A Christmas Carol`," focusing on the relationships and dynamics among characters such as Ebenezer Scrooge, the Cratchit family, and the various spirits. The analysis will explore themes of redemption, familial bonds, and social responsibility, providing insights into how these elements reflect the broader societal context of Dickens' time. The report will be used to inform educators, literary scholars, and enthusiasts about the significance of character interactions and thematic developments within the narrative, enhancing their understanding and appreciation of the text.. The content of this report includes an overview of the community's key entities and relationships.

Report Structure

The report should include the following sections:

- TITLE: community's name that represents its key entities - title should be short but specific. When possible, include representative named entities in the title.
- SUMMARY: An executive summary of the community's overall structure, how its entities are related to each other, and significant points associated with its entities.
- REPORT RATING: A float score between 0-10 that represents the relevance of the text to literary analysis, character development, thematic exploration, and narrative structure, with 1 being trivial or irrelevant and 10 being highly significant, profound, and impactful to the understanding of the text and its implications within the literary canon.
- RATING EXPLANATION: Give a single sentence explanation of the rating.
- DETAILED FINDINGS: A list of 5-10 key insights about the community. Each insight should have a short summary followed by multiple paragraphs of explanatory text grounded according to the grounding rules below. Be comprehensive.

Creating custom prompt files

For the most advanced users with niche datasets, any of the GraphRAG system prompts can be customized.

Each prompt has several {tokens} that can be integrated within your custom prompts – see public docs for more.

-Goal-

Given a text document that is potentially relevant to this activity and a list of entity types, identify all entities of those types from the text and all relationships among the identified entities.

-Steps-

1. Identify all entities. For each identified entity, extract the following information:

- entity_name: Name of the entity, capitalized
- entity_type: One of the following types: [{entity_types}]
- entity_description: **Comprehensive description of the entity's attributes and activities**

Format each entity as

("entity"{tuple_delimiter}<entity_name>{tuple_delimiter}<entity_type>{tuple_delimiter}<entity_description>{tuple_delimiter}<relationship_description>{tuple_delimiter}<relationship_strength>{tuple_delimiter}<relationship_strength>")

2. From the entities identified in step 1, identify all pairs of (source_entity, target_entity) that are ***clearly related*** to each other.

For each pair of related entities, extract the following information:

- source_entity: name of the source entity, as identified in step 1
- target_entity: name of the target entity, as identified in step 1
- relationship_description: **explanation as to why you think the source entity and the target entity are related to each other**
- relationship_strength: **a numeric score indicating strength of the relationship between the source entity and target entity**

Format each relationship as

("relationship"{tuple_delimiter}<source_entity>{tuple_delimiter}<target_entity>{tuple_delimiter}<relationship_description>{tuple_delimiter}<relationship_strength>{tuple_delimiter}<relationship_strength>")

3. Return output in English as a single list of all the entities and relationships identified in steps 1 and 2. Use ****{record_delimiter}**** as the list delimiter.

4. When finished, output {completion_delimiter}

#####

-Examples-

#####

Example 1:

Entity_types: ORGANIZATION,PERSON

Text:

The Verdantis's Central Institution is scheduled to meet on Monday and Thursday, with the institution planning to release its latest policy decision on Thursday at 1:30 p.m. PDT, followed by a press conference where Central Institution Chair Martin Smith will take questions. Investors expect the Market Strategy Committee to hold its benchmark interest rate steady in a range of 3.5%-3.75%.

#####

Output:

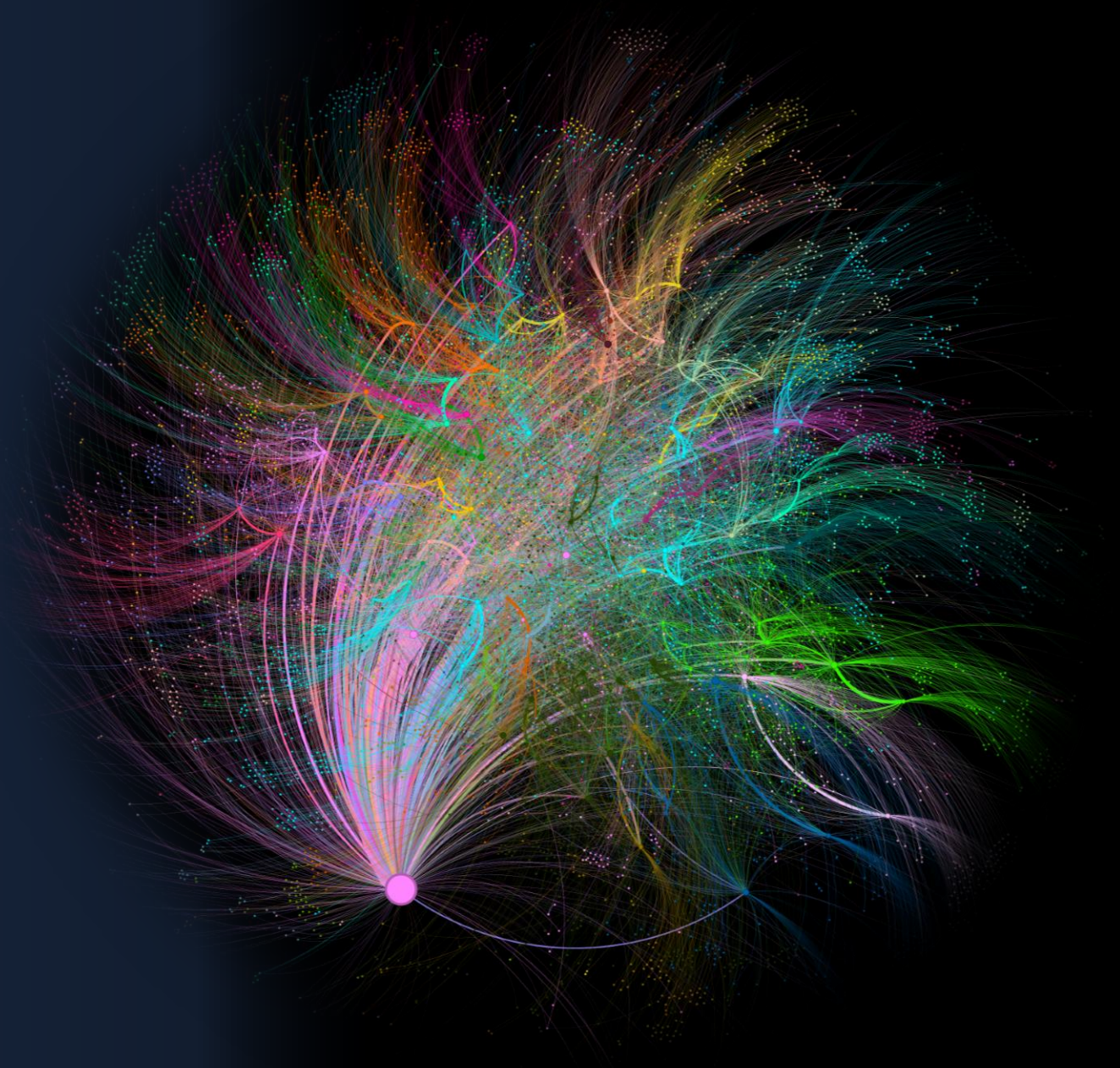
Section Summary

GraphRAG uses prompt tuning to emulate human domain expertise.

In addition to GraphRAG's default prompts, it provides an auto and manual tuning options.

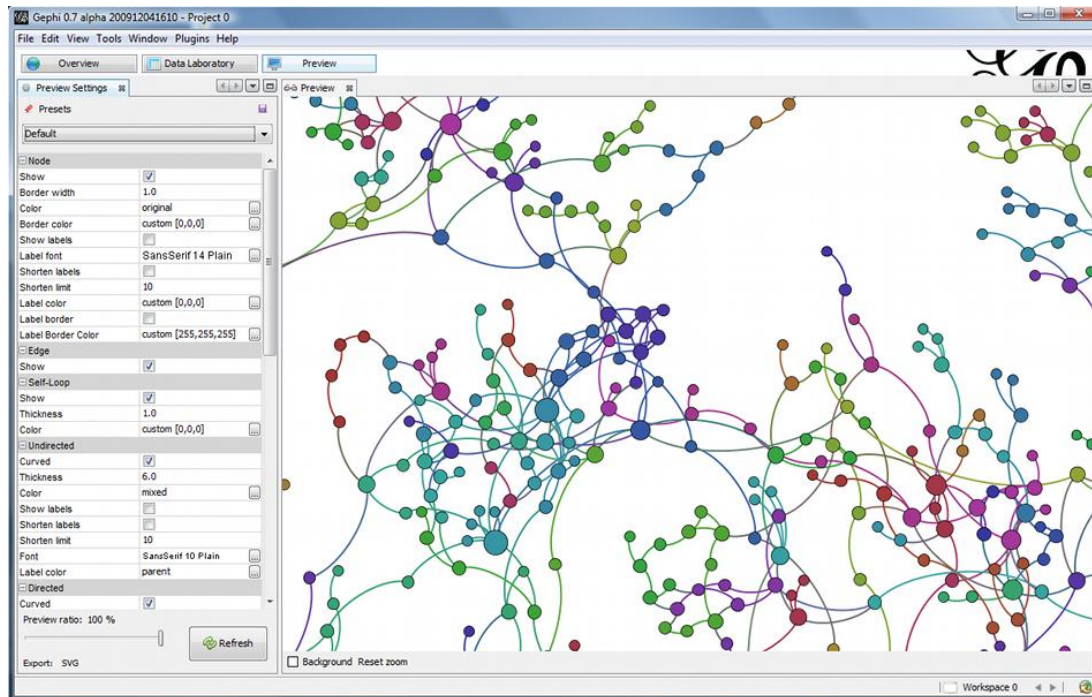
GraphRAG Auto-tuning uses a small sample from your data corpus to discover the data domain and craft custom prompts for you.

The system prompts are text files and can be manually edited to meet your specific needs.



Exploring the Knowledge Graph

What is Gephi

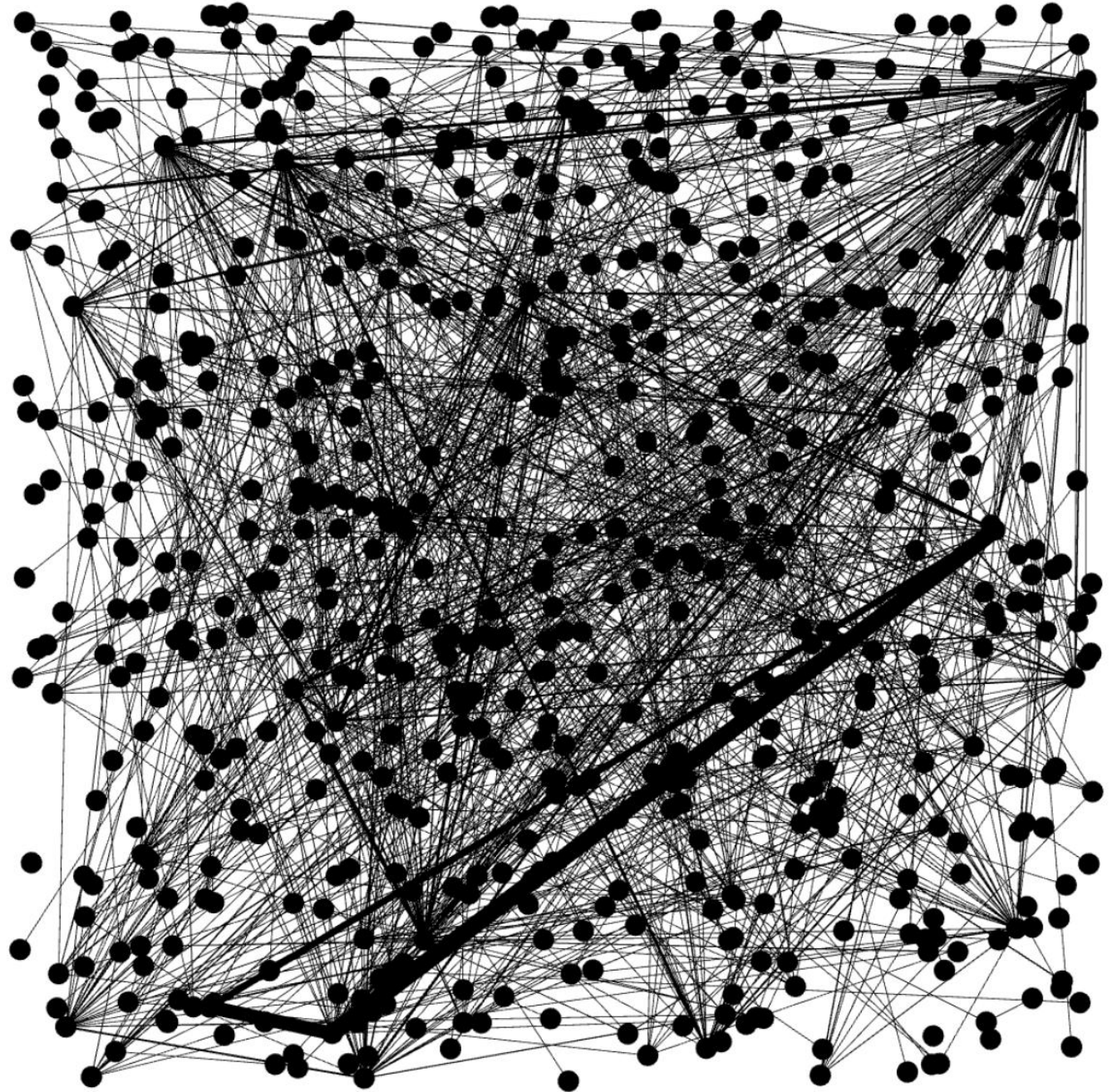


Gephi is a leading visualization and exploration software for all kinds of graphs and networks.

Gephi is open-source and free.

Open the Graph in Gephi

1. Install and open Gephi
2. Navigate to the [output](#) folder containing the various parquet files.
3. Import the [graph.graphml](#) file into Gephi. This will result in a plain view of the undirected graph nodes and edges.



Run graph statistics, including the Leiden Algorithm

Network Overview	
Average Degree	Run ?
Avg. Weighted Degree	Run ?
Network Diameter	Run ?
Graph Density	Run ?
HITS	Run ?
PageRank	Run ?
Connected Components	Run ?
Leiden algorithm	Run ?

In the Statistics tab on the right, Run the **Average Degree** and **Leiden Algorithm**.

Leiden setting adjustments:

- **Quality function:** Modularity
- **Resolution:** 1

Color the graph by clusters & resize nodes by degree



1. Click on Nodes, then select Partition, and p
2. Click the color palette icon located in the upper right corner.
3. From the dropdown menu, select Cluster.
4. Click on the Palette link and then choose Generate.
5. Deselect Limit number of colors, hit Generate, and then press Ok.
6. Finally, click Apply to apply the colors to the graph. This will color the graph according to the partitions identified by Leiden.



1. Click on Nodes, then select Ranking located in the upper left corner.
2. Click the Size icon found in the upper right corner.
3. Select Degree and configure:
4. **Min: 10**
5. **Max: 150**
6. Click Apply.

Layout the Graph using OpenOrd and ForceAtlas2

Layout ×

OpenOrd

Run

Stages

Liquid (%)	50
Expansion (%)	50
Cooldown (%)	0
Crunch (%)	0
Simmer (%)	0

1. In the Layout tab in the lower left, select OpenORD.
2. Set **Liquid and Expansion stages to 50**, and everything else to 0.
3. Click Run and monitor the progress.

1. Select ForceAtlas 2
2. Adjust settings: **Scaling 15, Check Dissuade Hubs, Uncheck LinLog mode, Check Prevent Overlap.**
3. Click Run and wait for Graph to settle. Click Stop.

ForceAtlas 2

Run

Threads

Threads number 7

Performance

Tolerance (speed) 1.0

Approximate Repulsion ☐

Approximation 1.2

Tuning

Scaling 15

Stronger Gravity ☐

Gravity 1.0

Behavior Alternatives

Dissuade Hubs ☒

LinLog mode ☐

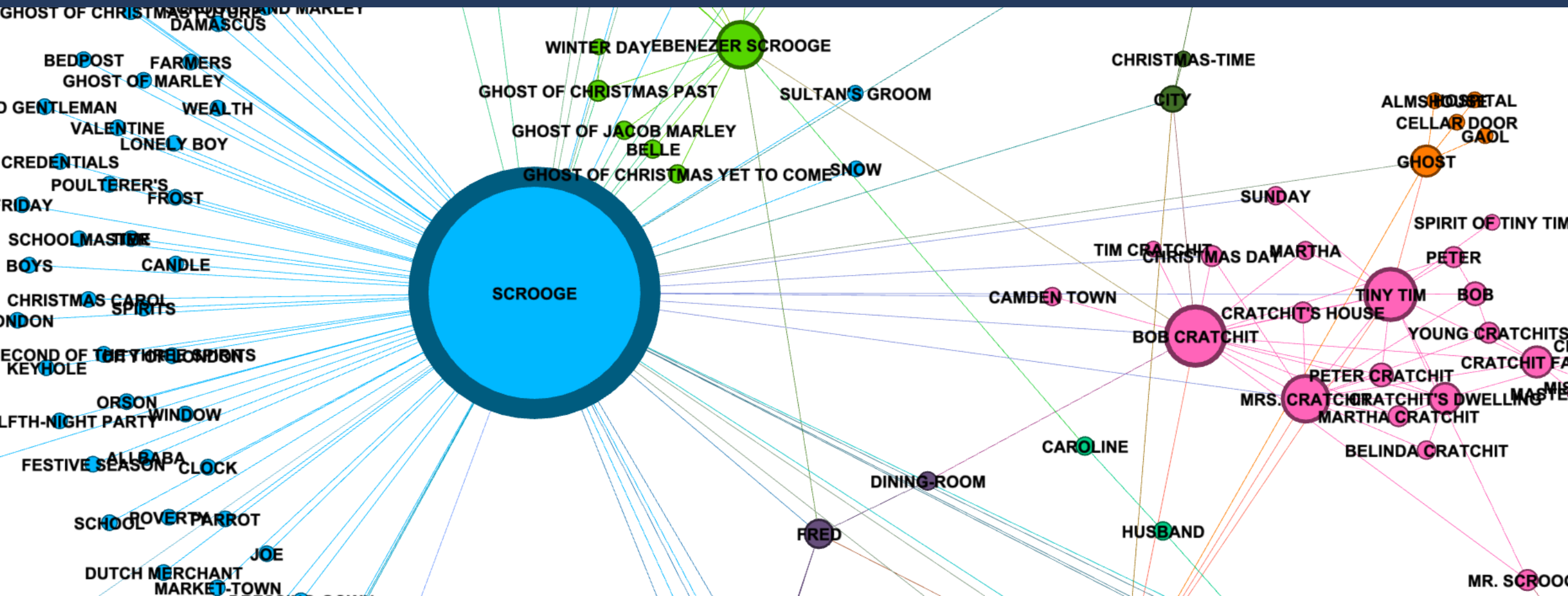
Prevent Overlap ☒

Edge Weight Influence 1.0

Normalize edge weights ☐

Inverted edge weights ☐

Explore!



Section Summary

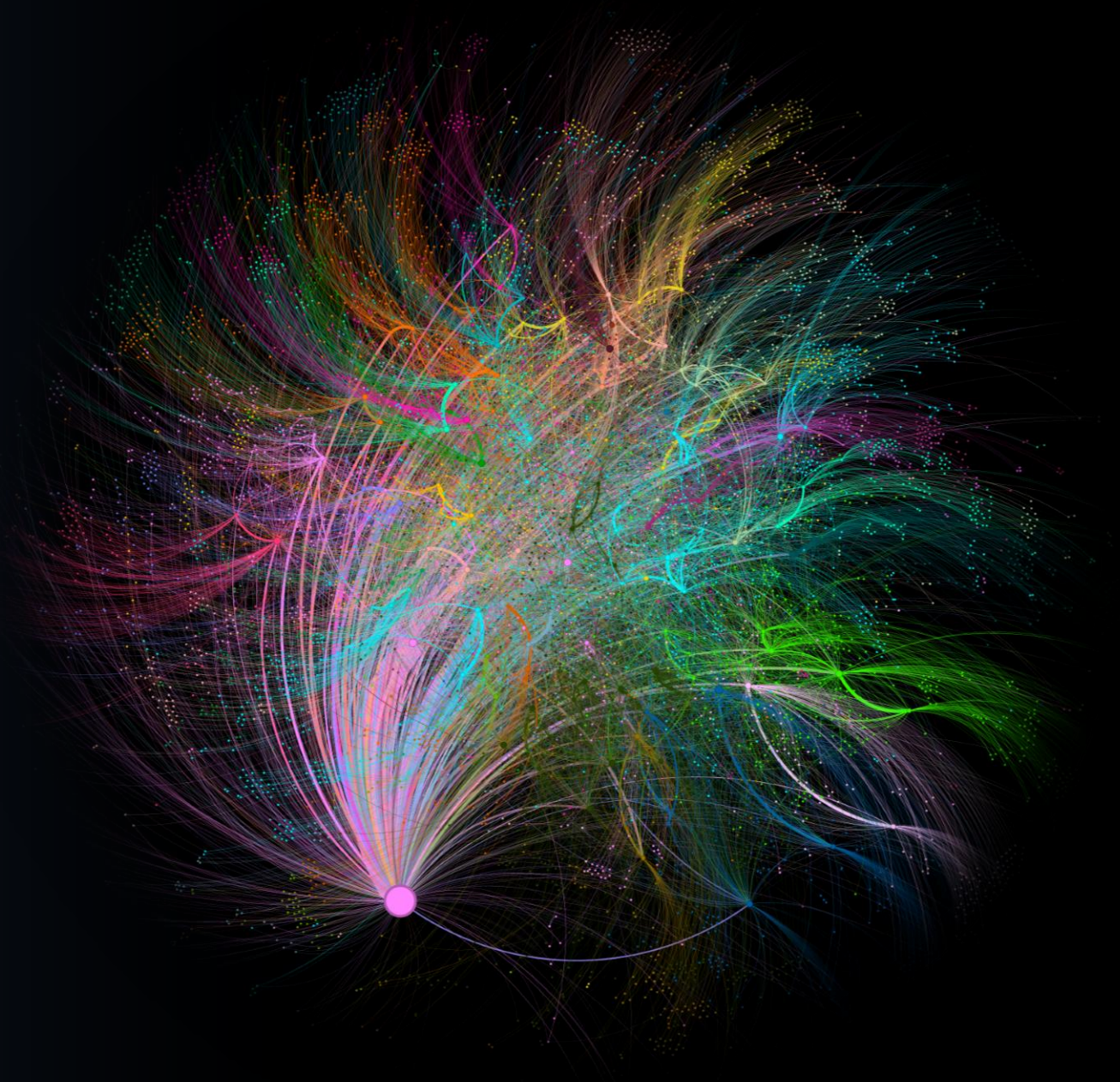
Gephi is a free and open-source tool (Gephi.org) that we recommend for exploring GraphRAG's knowledge graph.

Configure GraphRAG to output a `graphml` file in the `settings.yaml` file by changing `snapshots.graphml` to `true`. This must be done *prior* to indexing.

When first opening a GraphRAG `graphml` file in Gephi, it will show lots of connected black dots. You must do some manual steps to see the formed knowledge graph.

These steps include running the `Leiden clustering` algorithm, setting node size and assigning cluster colors.

Gephi is very powerful, so consider these tips a jumping off point towards your own explorations and preferences!



GraphRAG Accelerator for Azure

<https://github.com/Azure-Samples/graphrag-accelerator>

Lab – GraphRAG

Lab Description

- Simple end-to-end example for using the GraphRAG system, using the install from pypi option.